AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 30: Symmetry

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1 Keywords

- Inductive Bias, Symmetry, Invariance, Equivariance, Group theory, Deep Sets, Transformers

2 Main Points

2.1 Inductive Bias

- Learning machines are not universal approximators, but rather have a propensity to learn a specific set of rules
- This is known as their *inductive bias*
- We want to choose the learning machine so the inductive bias reflects the properties of the problem
- In almost all problems we use some smoothness property to select slowly varying functions

2.2 Invariances

- For many problems there are symmetries that you want your learning machine to respect
- For example in image classification we consider the class of an object to be invariant to its position in an image
- Learning machines whose outputs are invariant to translations have a much smaller set of functions to learn and therefore are much less likely to overfit
- CNNs are equivariant in that if we translate the input the output will be translated in the same way
- Building classifiers using layers of CNN builds in a lot of translational invariance
- Ideally we want a classifier system to also be scale and rotation invariant, although we don't know how to do this well
- Sometimes we use pyramid systems where we rescale the image and run the same operations on all scales
- However, in many datasets the foreground object is rescaled to fill most of the image with smaller object treated as the background so scale invariance is not necessarily required
- Often we try to learn invariance by augmenting the training set by artificially transforming the image (e.g. by reflecting, cropping, shearing and rescaling an image)
- Data augmentation usually gives a very significant improvement in performance
- We can sometimes achieve invariance by suitably normalising our data. Again when done right this can give a significant improvement in performance

2.3 Group Theory

- The mathematical language of symmetry is group theory
- You should know group theory exists, but I'm not going to examine you on details of group theory
- This considers a set of objects (we can think of these as transformations)
- Together with a binary operation we can think of as composition
- That is if $a, b \in \mathcal{G}$ are two transformations the $a \cdot b$ can be interpreted as applying transformation b followed by transformation a
- For the set, $\mathcal{G}$, and binary operation, "$\cdot$" to form a group they must satisfy four conditions
  1. Closure: if $a, b \in \mathcal{G}$ the $a \cdot b \in \mathcal{G}$
  2. Associativity: for all $a, b, c \in \mathcal{G}$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
  3. Existence of an identity element where $e \cdot a = a \cdot e = a$
  4. Every element, $a \in \mathcal{G}$ has an inverse such that $a^{-1} \cdot a = a \cdot a^{-1} = e$
- (Group theory doesn't really care if the elements of the group are transformations, just that the axioms are satisfied)
- An example is the cyclic group C_2
  - We can think of this as tossing a coin with two actions
    - flip the coin: f
    - leave the coin alone: e
  - These form a group as
    1. Closure: $f \cdot e = f$, $f \cdot f = e$, $e \cdot e = e$ and $e \cdot f = f$ (if you don't believe this get out a coin and try. The first identity is don't do anything then flip the coin is the same as just flipping the coin)
    2. Associativity: this follows automatically from the operation being a composition (we will see this in innerProduct lecture where we look at matrix multiplication). You can also check rather boringly, e,g, $(e \cdot f) \cdot f = f \cdot f = e$ while $e \cdot (f \cdot f) = e \cdot e = e$ thus $(e \cdot f) \cdot f = e \cdot (f \cdot f)$. You can try this for all triples
    3. e is clearly the identity
    4. $e^{-1} = e$ and $f^{-1} = f$
  - The cyclic group C_3 corresponds to the rotations of an equilateral triangle maps the triangle to itself. While C_n are the rotations that keep an n sided polygon with identical sides onto itself
  - The *four group* is a second group of size 4 (besides C_4) that can be thought of as describing the flipping around a vertical and horizontal axis a rectangle, or rotating it by 180 degress around its centre of mass. We can describe the group in terms of a *multiplication table* where the first row and first column corresponds to the elements a_i and the ith row and j^{th} column corresponds to the multiplication $a_j \cdot a_i$

	e	v	h	r
e	e	v	h	r
v	v	e	r	h
h	h	r	e	v
r	r	h	v	e

    - This is an "Abelian group" meaning that for any elements a, b we have $a \cdot b = b \cdot a$

- The group of permutations of n objects forms a group of size $n!$ called the symmetric group S_n
  - This is a group as the composition of a permutation is a permutation
  - There is an identity (doing nothing)
  - For every element there is an inverse permutation (which is a permutation)
  - For $n > 2$ (where $S_2 = C_2$) the group is non-Abelian meaning that there are permutations p_i and p_j where $p_i \cdot p_j \neq p_j \cdot p_i$
  - Despite this composing permutations is associative
  - Permutations have a lot of structure and S_n is very well studied
- There exists groups with an infinite number of elements. The set of integers with the operation of addition form a group with identity, 0, and the inverse of n is $-n$
  - You can think of these describing an infinite set of discrete shifts
  - You first started studying this group when you first learn to count
- The set of rational numbers under addition also forms a group
- The reals under addition form a group describing any one dimensional translation
- The set of reals excluding 0 forms a group under multiplication with an identity 1 and where the inverse of x is $1/x$
  - These can be thought of as describing scale transformations
- Groups can be interpreted in different ways with their elements having different names (e.g. one, two, three, $\cdots$ or un, deux, trois, $\cdots$). The permutation of two elements has the same structure (is isomorphic to) flipping a coin. What counts is the structure of the multiplication table (up to re-ordering the elements)

2.3.1 Lie Groups

- When we consider continuous transformation in high dimensional space then we find a set of very interesting (smooth) groups known as Lie groups
- These describe things like general translational invariance, rotational invariance, relativistic invariance
- The set of transformations can be represented as matrices, with the composition operator being matrix multiplication
- Different groups correspond to different sets of matrices.
- A well known group is the set of $n \times n$ orthogonal matrices $O(n)$
- By definition of orthogonal matrices $\mathbf{M}^\mathsf{T}\mathbf{M} = \mathbf{M}\,\mathbf{M}^\mathsf{T} = \mathbf{I}$
- These correspond to rotations and reflections
- Note that, in general, $\det(\mathbf{M}^\mathsf{T}) = \det(\mathbf{M})$ and $\det(\mathbf{A}\,\mathbf{B}) = \det(\mathbf{A}) \times \det \mathbf{B}$
- As a consequence $\det(\mathbf{M}^\mathsf{T}\mathbf{M}) = \det(\mathbf{I}) = 1$ so that $\det(\mathbf{M})^2 = 1$ or $\det(\mathbf{M}) = \pm 1$
- If $\det(\mathbf{M}) = -1$ then the transformation involves a reflection
- The special orthogonal group, $SO(n)$, corresponds to a subgroup of matrices in $O(n)$ with $\det(\mathbf{M}) = 1$. These correspond to rotations only
- The group $SO(2)$ corresponds to 2-d rotations where the elements of the group are matrices

$$\begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix}$$

2.4 Symmetry in Machine Learning

- **Invariance by averaging.** For any function f and finite group G of transformations, $\bar{f}(\boldsymbol{x}) = \frac{1}{|G|}\sum_{g\in G} f(g\cdot\boldsymbol{x})$ is invariant: for $h \in G$,

$$\bar{f}(h\cdot\boldsymbol{x}) = \frac{1}{|G|}\sum_{g\in G} f(g\cdot h\cdot\boldsymbol{x}) = \frac{1}{|G|}\sum_{g'\in G} f(g'\cdot\boldsymbol{x}) = \bar{f}(\boldsymbol{x})$$

  because $g \mapsto g' = g\cdot h$ is a one-to-one map of G onto itself (by closure, and because $g = g'\cdot h^{-1}$). The group axioms are exactly what is needed. Data augmentation approximates this average during training; *test-time augmentation* computes it at prediction time

- **Equivariance and convolution.** A map is equivariant if $f(g\cdot\boldsymbol{x}) = g\cdot f(\boldsymbol{x})$. Convolution commutes with shifts, and the linear maps that commute with all (circular) shifts are exactly the convolutions. A 3×3 filter has 9 weights against n^4 for a dense layer on an $n\times n$ image: building in the symmetry removes parameters and with them variance

- **Permutation symmetry.** For a set of inputs, any function $f(\{\boldsymbol{x}_1,\dots,\boldsymbol{x}_n\}) = \rho\big(\sum_i \boldsymbol{\phi}(\boldsymbol{x}_i)\big)$ is invariant to their order (*Deep Sets*, Zaheer et al., 2017). Graph neural networks update each node from a sum over its neighbours, so they are equivariant to relabelling the nodes

- **Transformers.** In self-attention each token computes $\boldsymbol{q}_i$, $\boldsymbol{k}_i$, $\boldsymbol{v}_i$ from its own input and outputs

$$\boldsymbol{y}_i = \sum_j \mathrm{softmax}_j\Big(\frac{\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{k}_j}{\sqrt{d}}\Big)\boldsymbol{v}_j$$

  If the tokens are permuted, each output is computed from the same set of terms, so the outputs are permuted in the same way: self-attention is permutation-equivariant. It therefore cannot see word order, which is why transformers add *positional encodings*

- **Rotations and molecules.** The energy of a molecule is invariant to rotations and translations, and the forces on its atoms are equivariant. Equivariant networks build this in and are now standard for molecules and materials; this area is called *geometric deep learning*

2.5 Looking Back

Represent	vectors, inner products, eigenvectors, SVD, PCA
Fit	derivatives, gradient descent, Newton, SGD, Adam
Constrain	Lagrange multipliers, convexity, duality, SVMs, kernels
Be uncertain	probability, Bayes, Gaussians, GPs, MCMC
Compress	entropy, information, KL, optimal transport
Generalise	bias and variance, regularisation, symmetry

- The same ideas recur: the SVD in PCA, ridge regression and double descent; Lagrange multipliers in SVMs, maximum entropy and optimal transport

- Machine learning works when the model's structure matches the structure of the world. Priors, regularisers and symmetries are three ways of saying what that structure is, and mathematics is the language for saying it precisely

3 Exercises

3.1 Averaging over flips

- Let $G = \{e, r\}$ where r reflects an image left to right. Show that G is a group, and that $\bar{f}(\boldsymbol{x}) = \frac{1}{2}\big(f(\boldsymbol{x}) + f(r\cdot\boldsymbol{x})\big)$ is invariant under r.

3.2 Self-attention is permutation-equivariant

- Write the inputs as the rows of a matrix $\mathbf{X}$ and a permutation as a permutation matrix $\mathbf{P}$. With $\mathbf{Q} = \mathbf{X}\mathbf{W}_q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_k$, $\mathbf{V} = \mathbf{X}\mathbf{W}_v$ and the softmax taken along each row, show that permuting the inputs permutes the outputs.

4 Experiments

4.1 Checking the symmetries

- Check numerically that self-attention is permutation-equivariant and that averaging over the flip group gives invariance

```
import numpy as np

rng = np.random.default_rng(0)
n, d = 5, 4                                  # 5 tokens, dimension 4
X = rng.standard_normal((n, d))
Wq, Wk, Wv = (rng.standard_normal((d, d)) for _ in range(3))


def attention(X):
    Q, K, V = X @ Wq, X @ Wk, X @ Wv
    A = np.exp(Q @ K.T / np.sqrt(d))
    A /= A.sum(1, keepdims=True)             # softmax over j
    return A @ V


perm = rng.permutation(n)
print(np.allclose(attention(X[perm]), attention(X)[perm]))   # equivariant

# invariance by averaging over the group {identity, left-right flip}
img = rng.standard_normal((6, 6))
f = lambda x: (x * np.arange(36).reshape(6, 6)).sum()      # not invariant
fbar = lambda x: (f(x) + f(x[:, ::-1])) / 2
print(f(img), f(img[:, ::-1]))
print(fbar(img), fbar(img[:, ::-1]))
```

- This prints `True`; f gives different values for the image and its mirror image (-27.1 and -6.5), while $\bar{f}$ gives the same value (-16.8) for both

5 Answers

5.1 Averaging over flips

- $r \cdot r = e$, so the set is closed, e is the identity, each element is its own inverse, and composition is associative: it is the group C_2
- $\bar{f}(r \cdot \boldsymbol{x}) = \frac{1}{2}\big(f(r \cdot \boldsymbol{x}) + f(r \cdot r \cdot \boldsymbol{x})\big) = \frac{1}{2}\big(f(r \cdot \boldsymbol{x}) + f(\boldsymbol{x})\big) = \bar{f}(\boldsymbol{x})$

5.2 Self-attention is permutation-equivariant

- Permuting the inputs gives $\mathbf{PX}$, so $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ become $\mathbf{PQ}$, $\mathbf{PK}$, $\mathbf{PV}$ and the score matrix becomes $\mathbf{P}\mathbf{Q}\mathbf{K}^{\mathsf{T}}\mathbf{P}^{\mathsf{T}}/\sqrt{d}$: its rows and columns are permuted in the same way

- The softmax acts on each row separately and a permutation of a row's entries just permutes its softmax, so the attention matrix becomes $\mathbf{PAP}^\mathsf{T}$
- The output is $\mathbf{PAP}^\mathsf{T}\mathbf{PV} = \mathbf{PAV}$, using $\mathbf{P}^\mathsf{T}\mathbf{P} = \mathbf{I}$: the outputs are permuted